La Scuola d'Italia / Dr. Huson / IB Math 12
11 November 2026

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3.5 Pre-Quiz: Trigonometry Review

A calculator is allowed. Give answers correct to three significant figures, unless they are exact. Angles are in radians unless they are given in degrees. Diagrams are not to scale.

1. [Maximum mark: 11]

   Let $\cos\theta = \frac{1}{4}$, where $\frac{3\pi}{2} < \theta < 2\pi$.

   (a) Find the exact value of $\sin\theta$. [3]

   (b) Find the exact value of $\sin 2\theta$. [2]

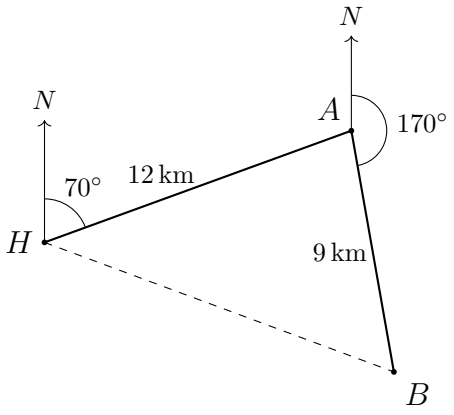
   (c) Find the exact value of $\cos 2\theta$. [2]

   (d) Show that $\frac{\sin 2x}{1+\cos 2x} = \tan x$, for $\cos x \neq 0$. [3]

   (e) Hence, or otherwise, find the exact value of $\tan\theta$. [1]

2. [Maximum mark: 12]

A ship leaves harbour H and sails 12 km on a bearing of 070° to point A. It then sails 9 km on a bearing of 170° to point B.



(a) Show that $H\hat{A}B = 80°$. [2]

(b) Find the distance HB. [3]

(c) Find the bearing of B from H. [4]

(d) Write down the bearing of H from B. [1]

(e) Find the area of triangle HAB. [2]

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3. [Maximum mark: 11]

Give your answers as exact multiples of π.

(a) Solve the equation $\sin 2x = \sin x$ for $0 \leq x \leq 2\pi$. [5]

(b) (i) Show that $\cos 2x + 3\cos x + 2 = 2\cos^2 x + 3\cos x + 1$. [2]

(ii) Hence solve the equation $\cos 2x + 3\cos x + 2 = 0$ for $0 \leq x \leq 2\pi$. [4]

4. [Maximum mark: 13]

Let $f(x) = 6\sin x \cos x + 1$, for $0 \leq x \leq \pi$.

(a) Show that $f(x) = 3\sin 2x + 1$. [2]

(b) Write down the period of f. [1]

(c) Find the range of f. [2]

(d) Sketch the graph of $y = f(x)$ on the grid below. [3]

(e) Solve the equation $f(x) = 2.5$. Give your answers as exact multiples of π. [3]

(f) Find the values of k for which the equation $f(x) = k$ has exactly two solutions. [2]

